

# Presentations, Seminar Talks, & Conferences

Diljit Singh

## Research Presentations

1. Combinatorial Models of Quantum Matrices  
*Joint Mathematics Meetings 2018*  
(Session on Structure & Representations of Hopf Algebras)
2. Laplacians on Projective Fractals  
*Young Mathematicians Conference, 2017*
3. The Intersection Number of Certain Transformation Digraphs  
*Macaulay Honors College Research Expo, 2017*
4. Spectral Results on Projective Fractals  
*Cornell University, Undergraduate Research Forum, 2016*
5. **Poster:** Quantum Determinant Identities via Cauchon Diagrams  
*University of California Santa Barbara, Bren Hall Poster Session, 2017*

## Graduate Seminar Talks

6. An Introduction to Additive Number Theory **1 Hour**  
*Graduate Student Seminar, Stony Brook University, 2019*
7. Atiyah's "On Analytic Surfaces with Double Points" **1 Hour**  
*Graduate Algebraic Geometry, Stony Brook University, 2019*
8. Infinite periodic groups and Burnside problems **1 Hour**  
*Groups & Semigroups with Applications to Theoretical Computer Science, 2018*
9. Regular Local Rings are Factorial **2 Hours**  
*Algebraic Geometry Lectures, CUNY Graduate Center, 2018*
10. Tveberg's Theorem (5 part seminar series) **5 Hours Total**  
*Discrete Geometry Student Seminar, CUNY Graduate Center, 2018*
11. Hard Walks on Good Sequences **45 Minutes**  
*Fortnight Research Dinners, Brookdale, 2018*
12. Proof of the Witness Theorem **45 Minutes**  
*Topics in Computation, CUNY Graduate Center, 2017*
13. Kelly's Resolution of The Sylvester–Gallai Problem of J. P. Serre **1 Hour**  
*Discrete Geometry Student Seminar, CUNY Graduate Center, 2017*
14. The Golod–Shafarevich theorem **2 Hours**  
*Number Theory Student Seminar, CUNY Graduate Center, 2017*

15. On Hirzebruch's "Arrangements of Lines and Algebraic Surfaces" **2 Hours**  
*Algebraic Geometry Lectures, CUNY Graduate Center, 2017*
16. Variations of the Sum-Product Problem **2 Hours**  
*Number Theory Student Seminar, CUNY Graduate Center, 2017*
17. Small Multiplicative Subgroups are not Sumsets **2 Hours**  
*Number Theory Student Seminar, CUNY Graduate Center, 2017*
18. Using Algebraic NT to Determine Rational Points on a Curve of Selmer **2 Hours**  
*Algebraic Geometry Lectures, CUNY Graduate Center, 2017*
19. Infinite Descent and  $E(\mathbb{Q})/2E(\mathbb{Q})$  **2 Hours**  
*Algebraic Geometry Lectures, CUNY Graduate Center, 2017*
20. Proof of the Weak Mordell-Weil Theorem **2 Hours**  
*Algebraic Geometry Lectures, CUNY Graduate Center, 2017*
21. Discrete Geometry Problem Session **1 Hour**  
*Discrete Geometry Student Seminar, CUNY Graduate Center, 2017*
22. The Crossing Number Inequality & the Szemerédi-Trotter Theorem **1 Hour**  
*Discrete Geometry Student Seminar, CUNY Graduate Center, 2017*
23. The Effective, Combinatorial, & Hilbert's Nullstellensatz **2 Hours**  
*Algebraic Geometry Lectures, CUNY Graduate Center, 2016*
24. Trivial (over  $\mathbb{Q}$ ) Curves of Genus 1 & Failure of the Hasse Principle **2 Hours**  
*Algebraic Geometry Lectures, CUNY Graduate Center, 2016*

## Undergraduate and High School Aimed Talks

25. The Polynomial Method in Olympiad Problems  
*Stony Brook University, 2019*
26. On (Some of) Hilbert's 23(+1) Problems  
*Macaulay Splash, 2017*
27. Applications of the Probabilistic and Polynomial Method to Graph Theory  
*Discrete Math Lecture, CUNY Hunter College, 2015*
28. On recent results in Intersection Digraph Theory (2 Parts)  
*Graph Theory Lectures, CUNY Hunter College, 2015*

## Summer Schools & Conferences

1. **Funded** Analytic & Combinatorial Number Theory: The Legacy of Ramanujan  
*University of Illinois Urbana-Champaign, Champaign, June. 6-9: 2019*
2. Combinatorial and Additive Number Theory (CANT)  
*CUNY Graduate Center, NYC, May 22-25: 2018*
3. **Funded** Arizona Winter School 2018: Iwasawa Theory  
*Arizona State University, Tucson, March. 3-7: 2018*

4. **Funded** Joint Math Meetings 2018  
*San Diego, California, Jan. 10-13: 2018*
5. **Funded** AMS Short Course: Discrete Differential Geometry  
*San Diego, California, Jan. 8-9: 2018*
6. **Funded** SACNAS National Conference  
*Salt Lake City, Utah, Oct. 19-21: 2017*
7. **Funded** ICERM Modern Math Workshop  
*Brown University, Salt Lake City, Oct. 18-19: 2017*
8. **Funded** Young Mathematicians Conference  
*Ohio State University, Columbus, Aug. 11-14: 2017*
9. **Funded** Thematic Program on Kahler Geometry  
*University of Notre Dame, June 12-16: 2017*
10. **Funded** 3<sup>rd</sup> Annual Graduate Student Conference in Algebra, Geometry, & Topology  
*Temple University, Philadelphia, June 3-4: 2017*
11. Combinatorial and Additive Number Theory (CANT)  
*CUNY Graduate Center, NYC, May 23-26: 2017*
12. AMS Spring Eastern Sectional Meeting  
*Hunter College: NYC, May 6-7: 2017*
13. Mathfest  
*CUNY Graduate Center, NYC, March 10: 2017*
14. **Funded** Math Day  
*Stony Brook University, Long Island, Nov. 12: 2016*
15. Combinatorial and Additive Number Theory (CANT)  
*CUNY Graduate Center, NYC, May 24-27: 2016*
16. Manhattan Algebra Day  
*CUNY Graduate Center, NYC, Dec. 9: 2015*
17. International Conference on Harmonic Analysis and Applications (ICHAA)  
*CUNY Graduate Center, NYC, June 1-5: 2015*
18. Combinatorial and Additive Number Theory (CANT)  
*CUNY Graduate Center, NYC, May 19-22: 2015*
19. Mathfest  
*CUNY Graduate Center, NYC, Nov. 11: 2014*